

UNIVERSITY OF NEVADA, LAS VEGAS

Department of Civil and Environmental Engineering and Construction | Dr. Sohn’s Research Lab

TO: PhD Applicant Candidate

DEADLINE: September 25, 2026

SUBJECT: Technical Assessment: Time Series Forecasting & Computer Vision

PART 1: CAPITAL BIKESHARE DEMAND ANALYSIS & FORECASTING

1. Assessment Objective & Background

The primary objective of this technical assessment is to evaluate your data analytics, statistical reasoning, and predictive modeling capabilities using real-world micro-mobility data from Washington, D.C. Through this assessment, you will analyze how temporal patterns, weather conditions, and user types (casual vs. registered) influence urban bike-sharing demand, providing both statistical insights and robust predictive models.

2. Data Fields Description

Please download the dataset using the following link:
<https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>. The dataset provides hourly rental records spanning two years with the following attributes:

Field Name	Description & Values
datetime	Hourly date + timestamp (YYYY-MM-DD HH:00:00)
season	1 = spring, 2 = summer, 3 = fall, 4 = winter
holiday	Whether the day is considered a holiday (1 = holiday, 0 = non-holiday)
workingday	Whether the day is neither a weekend nor holiday (1 = working day, 0 = weekend/holiday)
weather	1: Clear, Few clouds, Partly cloudy 2: Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist 3: Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds 4: Heavy Rain + Ice Pallets + Thunderstorm + Mist, Snow + Fog
temp	Temperature in Celsius
atemp	Feels like temperature in Celsius
humidity	Relative humidity (%)
windspeed	Wind speed

casual	Number of non-registered user rentals initiated
registered	Number of registered user rentals initiated
count	Number of total rentals (casual + registered)

3. Assessment Tasks

Please conduct an investigation of the dataset and address the following three main tasks in your report and accompanying code:

Task 1: Descriptive Statistics & Exploratory Analysis

Explore the dataset to identify and explain noteworthy descriptive statistics, patterns, or anomalies. (For example, analyze rental frequencies across different time windows or weather conditions, and compare the behavioral dynamics between casual and registered users).

Task 2: Statistical & Regression Analysis

Conduct a regression analysis to examine the relationship between the independent variables (e.g., weather, temperature, workingday) and the dependent variable (e.g., count, casual, or registered). Clearly discuss your findings, variable selection logic, and any statistical issues identified (e.g., multicollinearity, non-linearity, count data distribution).

Task 3: Demand Forecasting & Predictive Modeling

You are provided with a complete dataset containing hourly rental records spanning a two-year horizon. In this task, you are required to perform the dataset partitioning yourself before training and evaluating your models:

- Training set: The first 20 days of each month (days 1 through 20).
- Test set: From the 21st through the end of each month.

Your objective is to partition the dataset accordingly and build a forecasting model to predict the total hourly rental count (count) for each hour in the evaluation windows. You must ensure that all features rely strictly on information available prior to each prediction period (preventing lookahead bias and temporal data leakage). In your pipeline, address:

- Your choice of evaluation metric(s)
- Your feature engineering strategy
- Your selected model architecture(s) and training setup
- Empirical results and performance discussion across the evaluation windows

PART 2: BUILDING FOOTPRINT SEGMENTATION

1. Dataset & Specifications

Please download the dataset using the following link: <https://project.inria.fr/aerialimagelabeling/>

Attribute	Details
Dataset Name	Inria Aerial Image Labeling Dataset
Task Category	Binary Semantic Segmentation (Building Footprints vs. Background)

Data Download Links	Official Site: https://project.inria.fr/aerialimagelabeling/ Kaggle: https://www.kaggle.com/dansbecker/inria-aerial-image-labeling-dataset
Dataset Specs	• Dimensions: 5000 x 5000 pixels per tile (0.3m spatial resolution RGB) • Coverage: Urban and suburban regions across various cities (Austin, Vienna, Chicago, Kitsap, Tyrol) • Mask Values: 255 = Building (Foreground), 0 = Non-Building (Background)

2. Assessment Question

Your objective is to design, implement, and evaluate an end-to-end binary semantic segmentation pipeline that accurately extracts building footprints from high-resolution aerial imagery. You are expected to independently reason through, execute, and justify every technical decision across your workflow.

You will design and code your own data partitioning scheme to split the dataset into training and evaluation sets, explicitly demonstrating how your partitioning logic prevents spatial data leakage (such as overlapping neighboring patches or geographic leakage across cities). Furthermore, you must select and justify your segmentation model architecture, formulate a loss function suited for binary masks that accounts for class imbalance and edge boundary precision, and choose evaluation metrics beyond simple pixel accuracy. Finally, you will present your model's empirical quantitative performance on the evaluation set alongside a qualitative visual assessment comparing the original aerial patch, ground-truth mask, and predicted mask, detailing both successful extractions and failure cases.

DELIVERABLES & SUBMISSION GUIDELINES

Please submit your completed technical assessment by the deadline via email or the designated submission portal. Your submission must include:

1. Executable Jupyter Notebook (.ipynb): Annotated, fully executable code demonstrating data loading, EDA, regression modeling, time-series forecasting pipeline (Part 1), as well as data partitioning, model training, evaluation metric computation, and predicted mask visualization for computer vision (Part 2).

2. Research Summary Report (.pdf): A concise technical memorandum presenting your statistical findings, regression tables, time-series predictive performance metrics (Part 1), along with your reasoning behind spatial data partitioning, model/loss choices, quantitative test results, and visual error analysis for semantic segmentation (Part 2).